Amitesh K Chaudhary

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Education

- **Indian Institute of Science (IISc)** Bengaluru, Karnataka
  PhD, Department of Mechanical Engineering; **CGPA**: 7.77 *2018 - 2025*
  Courses: *Turbulence processes and data-driven analysis, Computational Gas dynamics, Rheology of Complex fluids and Particulate materials, Convective Heat Transfer, Mathematics, Fluid Mechanics, Thermodynamics, Mathematical Analysis of Experimental Data, Data Analysis & Visualization*
- **National Institute of Technology (NIT)** Srinagar, Jammu and Kashmir
  Bachelor of Technology - Mechanical Engineering; **CGPA**: 8.133 *2011 - 2015*

Experience

- **Indian Institute of Science (IISc)** *April 2025 - Present*
  Research Associate
  - ◇ Experimental investigation of evaporating fuel jet-in-cross flow (JCF) using swirl combustor for gas turbine applications, using laser-based diagnostic measurements.
- **Indian Institute of Science (IISc)** *Oct. 2024 - April 2025*
  Research Associate (Provisional)
  - ◇ Experimental investigation on spray characteristics of vaporizers for gas turbine applications.
  - ◇ Laser-based diagnostic measurements used to determine atomization and evaporation efficiency of the combustor and atomizer in-line assembly.

Research Project

- **Rotary atomization of non-Newtonian liquids for Spray drying applications** : Research aims at studying and investigating the spray characteristics of viscoelastic polymer solutions using rotary atomization in spray drying conditions.
  - ◇ **Identification & characterization of polymer solution**:
    - ∗ Shear-thinning behavior of polymer solutions investigated using steady and dynamic shear rheometry on a controlled-stress MCR302 rheometer.
    - ∗ Extensional relaxation timescales and viscosity measurements obtained using CaBER-DoS technique.
  - ◇ **Rotary atomization experiments on insitu prepared water-based Xanthan gum solutions**:
    - ∗ Design of a shock-proof experimental facility for high-speed subsonic atomization studies.
    - ∗ Effect of polymer concentration from dilute-to-concentrated regime ranging from (0.1-4.0)wt.% at ambient conditions.
    - ∗ Effect of polymer solution temperature varied between 298 and 353 K.
    - ∗ Shadowgraphy and Particle/Droplet Image Analysis (PDIA) techniques used for determining the Sauter mean diameter (SMD) and droplet size distributions.
  - ◇ **Development of image processing technique to detect irregular-shaped ligaments** :
    - ∗ Edge detection and cluster analysis used to detect highly irregular-shaped ligaments and droplets.
    - ∗ Investigated the fraction of liquid solution in the form of ligaments and droplets.
    - ∗ Quantitative analysis of the atomization efficiency.
    - ∗ SMD and dropsize distributions were validated against results from imaging software DaVis (make: LaVision).
  - ◇ **Development of pendant single droplet experimental setup for droplet evaporation characterization** :
    - ∗ Experiments conducted on droplets pendant on a Ni-wire, continuously exposed to hot air flow.
    - ∗ Designed a novel closed-loop Arduino-based air temperature ($\pm$ 2 K) and humidity ($\pm$ 5 %) control system.
    - ∗ Effect of air temperature varied between 298 and 473 K.
    - ∗ Effect of air relative humidity varied between 1 and 50 %.
    - ∗ Pendant droplet diameter evolution determined using backlit droplet images.
    - ∗ Python-based images processing used to calculate the droplet evaporation timescales.
  - ◇ **Formulation of 1-D numerical model for single droplet evaporation with conjugate heat and mass transfer correlations**:
    - ∗ Two-stage single droplet 1D transient evaporation model solving the coupled equations of mass and energy conservation.
    - ∗ Incorporates Stefan-flow kinetics, conjugate heat and mass transfer correlations, solute thermal and physical properties, etc.

    * Sensitivity analysis conducted to study the effect of parameters like the hot air-flow temperature, air humidity, etc.
    * Incorporated material properties like density, heat capacity, thermal conductivity, etc as functions of temperature and solute concentration.
  - **Design of a portable, compact spray dryer prototype** :
    * Droplet sizes obtained from atomization and their evaporation timescales from single droplet study.
    * Designed a compact, portable spray dryer tailored for agricultural food products.

Skills

- Python
  - Image processing and cluster-analysis using OpenCV and scikit-learn libraries.
  - Data analysis and visualization using pandas, plotly, seaborn, numpy and matplotlib.
  - Machine learning using scikit-learn, CNN using TensorFlow and PyTorch.
- MATLAB
  - One dimensional numerical model solving coupled differential equations to predict the evaporation characteristics of a surrogate droplet in a given environment.
  - Image processing using Toolboxes.
- Experimental skills : Laser diagnostics, PIV, PTV, shadowgraphy, schlieren, PDIA, high-speed imaging, shear and extensional rheometry.
- CAD Modeling : SolidWorks, CATIA, Solid Edge, AutoDesk Fusion 360.
- Simulation : Computational Fluid Dynamics using Ansys Fluent, OpenFOAM and Comsol.
- Design of Experiments : A/B testing, ANOVA and hypothesis testing.
- Acquisition and analysis of data
- Familiarity with fire and combustion safety protocols
- Soft skills: Project & team management, Critical thinking, Resource & inventory management.

Achievements

- Prime Ministers' Research Fellow (PMRF) (2018 – 2023)
- Placed in the **Top 2%** (98.3 percentile) nationwide in the All India Entrance Examination (AIEEE - now IIT JEE), 2011.

Conferences & Publications

1. Amitesh K. Chaudhary, R.V. Ravikrishna; "Spray Characterization of Non-Newtonian liquid jets using a rotary atomizer", Institute for Liquid Atomization and Spray Systems (ILASS) Asia 2022, IIT Indore, India.
2. Amitesh K. Chaudhary, R.V. Ravikrishna; "Rotary atomization of viscoelastic liquid jets at high Ohnesorge numbers", XIXth International Congress on Rheology (ICR) 2023, Athens, Greece.
3. Amitesh K. Chaudhary, R.V. Ravikrishna; "Atomization of viscoelastic liquid jets using a rotary atomizer", Experimental Thermal and Fluid Science (to be submitted).
4. Amitesh K. Chaudhary, R.V. Ravikrishna; "Effect of temperature on the rheology and spray characteristics of polymer solutions", JNNFM (to be submitted).
5. Amitesh K. Chaudhary, R.V. Ravikrishna; "Experimental and numerical investigation on the effect of air temperature and humidity on single droplet evaporation", IJHMT (to be submitted).

Other Projects

- Machine learning
  - CNN-based Object recognition/classification of CIFAR10 and Fashion mnist datasets using TensorFlow and PyTorch.
- **Optimize customer waiting time at Srinagar International Airport using Queueing theory** : BTech final year project
  - Applied queuing theory using FIFO approach.
  - Determined the optimal number of servers required to reduce high customer waiting times due to stringent frisking measures, at different entry-exit points.
  - Recommended the use of a mobile server at the main Airport entrance.
  - Optimized server utilization and reduced customer waiting times at minimal cost.